CSS Advanced

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Advanced

- Images & Background Images
- Units & Sizing
- Centering Elements
- Text & Fonts
- Forms

Agenda

Image

- width & height
  - inside a inline element → referring to the image size
  - inside a block element → referring to the block element
- position & sizing is limited
  - hacky solution is to set a negative margin
  - so we often use a image container
- but all modern browser support
  - object-fit: **fill** | cover (ratio, clipped) | contain (ratio, full image)

Image Container

- use a image container ( )
  - for easier position & sizing
  - display: block | inline-block
  - vertical-align as needed
  - maybe add some box-shadow to it
- also good for responsiveness (next topic)

Background Image I

- background-image: url("image.webp")

- background-position: center | 50% 50%;
- background-position: left top | 0% 0%;

- background-size: 100%;
- background-size: auto 100%;
- background-size: cover;

- background-repeat: no-repeat;

Background Image II

- background: url("image.webp") left top/cover no-repeat
- but background images are bad for A11y and SEO
  - so really just use it for backgrounds, not for content!

Background Gradients

- background-image: linear-gradient(red, transparent);
- background-image: linear-gradient(to bottom, red, blue);
- background-image: linear-gradient(0deg, red, blue);

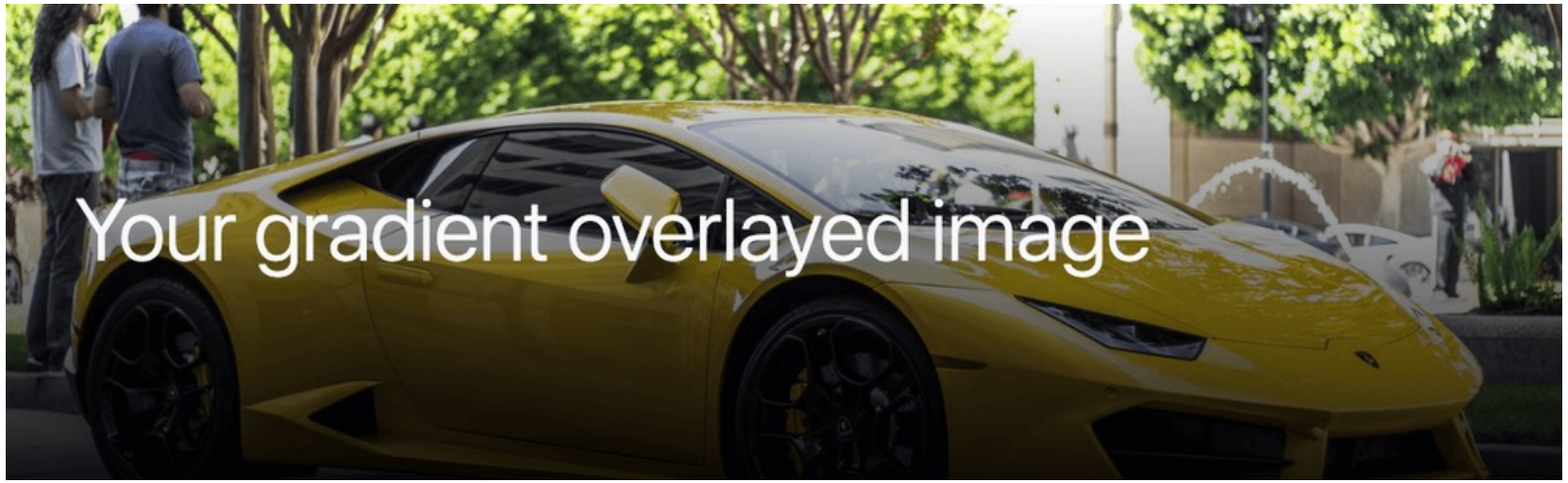
- background-image: radial-gradient(red, transparent)

- https://elementor.com/blog/gradients-in-web-design/
- https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_images/Using_CSS_gradients

Background Gradient & Image

```
.fancy-background {
  background: linear-gradient(180deg, rgba(0,0,0,0) 0%, rgba(0,0,0,0.90) 100%),
  url('/path/image.jpg');
  color: #fff;
}
```

```
<div class="fancy-background"> Your gradient overlayed image </div>
```



https://codingforeverybody.com/snippets/css-gradient-over-a-background-image

Filters

- applied on the background / image
- filter: blur(10px);
- filter: grayscale(10%);
- https://developer.mozilla.org/en-US/docs/Web/CSS/filter

SVGs

- fill:
  - like background-color:
- stroke:
  - like border-color:
- stroke-width:
  - like border-width:
- https://dev.../.../docs/Web/SVG/Tutorial/SVG_and_CSS

Units

- px
  - absolute, independent of user settings
- %
  - relative to containing block
    - position: static / relative → block ancestor
    - position: absolute → ancestor with position other than default (static)
    - position: fixed → viewport
- em & rem
  - relative to font-size
- vw & vh
  - relative to viewport size

Recommended Units For Sizing

- font-size
  - rem (em just for direct children)
- margin, padding
  - rem
- border
  - px
- width & height
  - % or vw/vh, sometimes px or rem
- top, right, bottom, left
  - %

Centering Elements

- margin: 0px auto; - for block elements
- text-align: center; - for inline elements
- grid or flex (we'll talk about them later in chapter 06)

```
.grid-container {
    display: grid; place-content: center;
}
.grid-element {
    grid-row: 1; grid-column: 1;
}
```

```
.flex-center {
    display: flex;
    justify-content: center;
    align-items: center;
}
```

https://www.joshwcomeau.com/css/center-a-div/#when-to-use-which-method-11

Text & Fonts

- Font-families
  - generic as fallback
  - Google Font!?
    - use a common font or
    - include the font in your NG project (@font-face) using .woff2 as format
- Font styling
  - font-size, font-weight, font-style
  - line-height, letter-spacing, white-space
  - text-decoration, text-shadow
  - white-space: no-wrap; overflow: hidden; text-overflow: ellipsis

Forms

- Form fields have default styling
  - which we tend to overwrite
- Provide a good form UX
  - Using :focus with border-color / outline
  - And .is-invalid with (red) border-color / outline

Best Practices

- position & size images containers/object-fit
- avoid background-images for content
- use apropriate units & centering
- include font as .woff2 via @font-face
- add user feedback (UX) to your forms

Tips

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Summary

Questions?

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